

6

A student investigated the distribution of plants in a heathland.

**Table 2** shows the number of plants he found in a sample area of 1 m<sup>2</sup>.

**Table 2**

| Species of plant | Number counted in<br>1 m <sup>2</sup> |
|------------------|---------------------------------------|
| Common heather   | 2                                     |
| Red fescue       | 14                                    |
| Vetch            | 2                                     |
| White clover     | 8                                     |

**0 6** . **1** What is the species richness of this sample?

[1 mark]

**0 6** . **2** Calculate the index of diversity of this sample. Show your working.

Use the following formula to calculate the index of diversity.

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

where  $N$  is the total number of organisms of all species  
and  $n$  is the total number of organisms of each species

[2 marks]

Index of diversity = \_\_\_\_\_

|   |   |   |   |
|---|---|---|---|
| 0 | 6 | . | 3 |
|---|---|---|---|

Suggest how this student would obtain data to give a more precise value for the index of diversity of this habitat.

**[2 marks]**

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**Turn over for the next question**